

Iacob Delcea

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SUMMARY

Junior Software Engineer focused on backend development with Python, FastAPI, PostgreSQL, Redis and SQLAlchemy. Experienced in building REST APIs, data processing workflows, integrations and database-backed applications. Background in networking, Linux and applied AI, with hands-on projects involving computer vision, graph databases and automation.

SKILLS

Backend: Python, FastAPI, REST APIs, SQLAlchemy, Pydantic

Databases: PostgreSQL, Oracle SQL, Redis, Neo4j, Cypher

AI/ML: PyTorch, Torchvision, OpenCV, Computer Vision, CNNs

Tools: GitHub, Docker Compose, Linux, Postman, Beautiful Soup

EDUCATION

University Politehnica of Bucharest (UNSTPB), Master's Degree in Advanced Digital Imaging Techniques Oct 2025 – Present

- **Status:** In progress (Year 1)
- **Focus:** Digital image processing, computer vision, advanced filtering, AI-based imaging

University Politehnica of Bucharest (UNSTPB), Bachelor's Degree in Networks and Software for Telecommunications (ETTI) Oct 2021 – July 2025

- Final grade: 9.8/10
- **Coursework:** Algorithms and Data Structures, Operating Systems, Computer Networks, Artificial Intelligence.

EXPERIENCE

Software Developer, Albocensa – Bucharest June 2024 – Present

- Developed and maintained backend services using Python, FastAPI, Redis, PostgreSQL and SQLAlchemy for internal business applications.
- Implemented REST API endpoints, database models, validation logic and caching mechanisms to support production workflows.
- Worked on secure B2Bi integrations, including data exchange, encryption and enterprise integration platforms.
- Troubleshoot issues in Linux environments and collaborated on integration flows involving networking protocols.

Laboratory Teaching Assistant (AI), University Politehnica of Bucharest Oct 2025 – Present

- Teaching laboratory sessions for the **Artificial Intelligence** course
- Covering core ML topics: **KNN, SVM, ANN, CNN**
- Designing lab exercises and evaluating student assignments
- Grading and assessing student performance

Technical IT Support Volunteer, ANCOM – Bucharest Sept 2024 – Oct 2024

- Solved technical issues encountered by conference delegates

PROJECTS

Game Recommendations System Neo4j

github.com/Iacob45/Game-Recommendations-System-Neo4J

- Developed a graph-based video game recommendation system using **Neo4j** and **FastAPI**, modeling users, games, genres, platforms, developers and publishers as connected graph entities.
- Implemented recommendation queries with **Cypher**, including user-based recommendations, similar games, similar users, graph statistics and basic performance measurements.
- Tools Used: Python, FastAPI, Neo4j, Cypher, Docker Compose, Postman

Beard Detection ResNet18

github.com/Iacob45/Beard-Detection-ResNet

- Developed a computer vision pipeline for multi-class facial hair classification using a pretrained **ResNet18** model and transfer learning.
- Built dataset preparation, training, testing and prediction workflows, including image preprocessing, face/lower-face cropping, data augmentation and confusion matrix analysis.
- Tools Used: Python, PyTorch, Torchvision, OpenCV, dlib, Pandas, Scikit-learn

Adaptive Median + DnCNN Image Denoising Pipeline

github.com/Iacob45/Denoising-Project

- Built an adaptive denoising pipeline combining a **selective median filter** (threshold-based correction) with a **DnCNN (residual learning)** model to remove remaining noise.
- Implemented a PyTorch training pipeline (custom dataset, Adam optimizer, ReduceLROnPlateau scheduler) and optimized image quality metrics such as **PSNR** and **SSIM**.
- Tools Used: Python, PyTorch, OpenCV, NumPy

AI-Powered Google Docs Parser

github.com/Iacob45/AI-Powered-Google-Docs-Parser

- Web application that extracts structured data from unstructured university timetables in Google Docs using AI and Regex-based logic.
- Tools Used: Python, Regex, GPT AI, FastAPI, Redis Stack, ReactJS, BeautifulSoup

XML, MDM, and XSD Parsing Tool (internal project @ Albocensa)

- Developed a custom tool for parsing XML, MDM and XSD files, extracting data to different formats.
- Tools Used: Python